

# Accessible Autonomous Vehicles: A Human-Computer Interaction Literature Review

Chikita Lengkong  
TU Ilmenau  
Ilmenau, Germany  
chikita.lengkong@tu-ilmenau.de

Cindy Mayas  
TU Ilmenau  
Ilmenau, Germany  
cindy.mayas@tu-ilmenau.de

Matthias Hirth  
TU Ilmenau  
Ilmenau, Germany  
matthias.hirth@tu-ilmenau.de

## Abstract

This systematic review investigates the intersection of Human-Computer Interaction, Accessibility, and Autonomous Vehicles by analyzing studies published between 2015 and 2025. A total of 32 studies were identified following Preferred Reporting Items for Systematic Reviews and Meta-Analyses guideline, a systematic screening process, across four academic databases, ACM Digital Library, Scopus, IEEE Xplore, and Web of Science. The subsequent qualitative coding process included the categories vehicle autonomy, target group, user tasks, input interaction mode, output feedback mode, and the usage-centered evaluation method. The results show that the research studies focus mainly on elderly people and people with visual restrictions. The majority of studies applied exploratory evaluation methods such as interviews, focus groups, surveys, or workshops.

## CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Accessibility.**

## Keywords

human-computer interaction, autonomous vehicles, accessibility

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## 1 Introduction

Autonomous Vehicles (AVs) are revolutionizing transportation, offering the potential to improve safety, efficiency, and mobility by minimizing human error, enhancing traffic flow, and optimizing travel through advanced navigation systems [40]. One of the most transformative aspects of AV adoption is its potential to increase accessibility for individuals who are unable to drive, including older adults and people with disabilities [23]. As AV technologies move toward widespread implementation in the real world, it becomes important to highlight how these systems accommodate the diverse needs of all users, especially users with special needs. In this paper,

the term "users with special needs" refers to individuals with disabilities as well as older adults, who often experience age-related challenges such as declines in vision, hearing, reaction time and cognitive processing.

The field of Human-Computer Interaction (HCI) provides a foundation for the development of accessible AVs. HCI research defines how AVs can effectively communicate information such as road and traffic conditions, navigation decisions, warnings, and status updates to passengers, pedestrians, and other road users via tactile feedback, visual interfaces, auditory indicators, and other external outputs. The integration of machine learning, wearable devices, gaze-tracking technology, real-time sensor data, Vehicle to Everything communication systems, and other advanced AV technologies plays a key role in the development of inclusive AVs. When combined with optimally designed ability, usability, and user experience features, these technologies contribute to broader acceptance among a wide range of users.

This literature review investigates the intersection of HCI and accessibility in the context of AVs, as illustrated in Figure 1. Current research often addresses HCI, accessibility, or AV each in isolation, which limits the development of inclusive AV solutions. By systematically analyzing the existing empirical studies, this study aims to identify critical research gaps in this integrated field. Addressing these gaps is essential for guiding the development of inclusive AV systems.

The following research question serves as the foundation for the investigation and analysis: **How do existing empirical studies on HCI in AVs address accessibility for users with special needs?** This research specifically focuses on the representation of target user groups, user tasks, interaction modality, and evaluation methods, aiming to provide a comprehensive framework that connects accessibility requirements in AV.

To achieve this, the paper is organized into the following sections: Section 2 reviews related works to establish the foundation of existing research. Section 3 details on the explanation of the systematic review methodology. Section 4 presents key findings of four categories: target user groups, user tasks, interaction modes, and evaluation approaches. These key findings are discussed in Section 5. Section 6 concludes the paper.

## 2 Related Works

HCI research in AV has increasingly addressed accessibility for users with special needs. Several literature reviews have examined assistive technologies and interaction modalities. However, prior works often analyzed isolated technologies or user groups, lacking a comprehensive synthesis of how AV systems accommodate different accessibility needs across varying levels of vehicle autonomy.

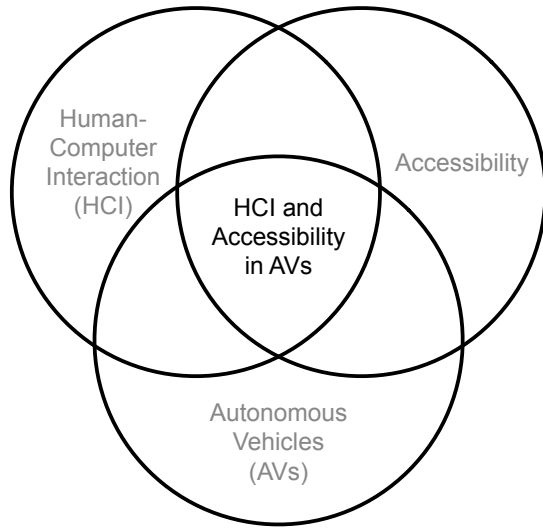


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**Figure 1: Thematic focus on the intersection of HCI, accessibility, and AV.**

The following subsections provide an overview of related studies, structured around key themes that correspond to the analytical categories in our qualitative analysis.

## 2.1 HCI Modalities in AV System

HCI modalities, such as touch, voice and gesture, play a crucial role in determining how users engage with AV systems. For users with special needs, the choice and design of these interaction modalities significantly affect usability and accessibility. However, existing studies have revealed critical challenges in this area, including the lack of modality options for different user capabilities and insufficient integration of assistive technologies [6].

Touch interfaces are currently the most common modality explored in AV system. However, this modality poses challenges for users with motor impairments, who may struggle with precise touch input [53]. As an alternative to touch interface, voice interaction has gained attention, particularly for users with visual or motor impairments, by enabling hands-free and eyes-free operation [16].

Furthermore, adaptive interaction mechanisms that personalize modality preferences based on user profiles have been suggested as a promising direction to address different accessibility needs [15, 30]. Personalization has been shown to improve accessibility for individuals with cognitive disabilities by adapting content and interaction methods to users' specific needs, abilities, and mobility contexts [42].

Designing interaction modalities is closely aligned with the specific tasks users need to perform. User tasks in AV involve a range of activities from booking a ride to exiting the vehicle, tasks that are relevant in AV systems. However, some tasks are more complex than the others, for example when it comes to emergency related tasks in comparison with passenger related tasks.

This approach is also relevant for older adults using AV systems, who often face age-related challenges such as declines in vision,

hearing, reaction time, and cognitive processing, which can affect their ability to perceive system feedback, respond to hazards, and interact with complex interfaces. Integrating personalized interfaces and targeted training can help mitigate risks and improve safety, making AV systems and technologies more accessible and accommodating to diverse user needs [8].

## 2.2 Evaluation Methods in Accessibility Research

Current research on accessibility in AV primarily uses exploratory methods such as interviews, focus groups, and surveys to understand user attitudes and expectations [3, 10, 19, 33]. While these methods help capture broad user perspectives, participatory design and expert evaluations enable the integration of accessibility needs in AV. Additionally, some research also focuses on technical system assessments without direct user input.

To understand the complexities of users-AV interactions, integrating exploratory evaluations with simulations or real-world testing is essential. Lab-based simulation and immersive technologies such as Augmented Reality (AR)/Virtual Reality (VR) have been used to simulate controlled driving scenarios for user training and interface evaluation [6, 31, 50], building trust and confidence without the risk associated with on-road testing [41, 49].

Expanding the role of immersive training in trust building, recent research has explored the potential of AVs to address social and safety challenges in public transportation. One study investigates the safety challenges faced by women in Bangladesh, emphasizing the prevalence of sexual harassment and assault in daily transit [5]. Using a mixed-method approach with 23 participants, the study combines a focus group and VR-supported co-design session to gather insights into participants' experiences and preferences for AV features that could improve safety in AVs.

While simulation-based methods provide a safe and flexible platform for early-stage evaluation, these methods may not fully capture the complexities and dynamic conditions of real-world driving. Field studies involving high-fidelity AV prototype are essential for validating how accessibility solutions perform in actual usage scenarios. However, such field implementations remain limited due to technical, infrastructure, legal, and social challenges [26, 44].

## 2.3 Accessibility in AVs for users with Special Needs

Prior research on accessibility and HCI in AVs has emphasized the importance of accommodating users with special needs. Table 1 summarizes three literature reviews that have examined the intersections of these topics. Chu and Huang [16] conducted a scoping review focusing on the role of voice assistants in AVs, particularly how they support users with visual and motor impairments through auditory feedback and task-relevant communication. This shows that most technologies are designed for users with visual or mobility challenges. Their study analyzed 35 publications between 2018 and 2023, using the PRISMA guideline to structure the review.

Another perspective is presented by Kettle and Lee [29], who conducted a systematic review of 31 articles published between 2012 and 2022, focusing on AR applications for vehicle-driver communication. Although their focus was on enhancing situational

**Table 1: Overview of literature review for research on HCI in accessibility for users with special needs.**

|                           | Chu & Huang et al. [16]                                                                                  | Kettle & Lee [29]                                                                                                                                                  | Bastola et al. [6]                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>              | The Intersection of Voice Assistants and Autonomous Vehicles: A Scoping Review                           | Augmented Reality for Vehicle-Driver Communication: A Systematic Review                                                                                            | Driving Toward Inclusion: A Systematic Review of AI-Powered Accessibility Enhancements for People With Disability in Autonomous Vehicles                                                                                    |
| <b>Timeframe</b>          | January 2018 - October 2023                                                                              | 2012 - 2022                                                                                                                                                        | 2017 - 2024                                                                                                                                                                                                                 |
| <b>Database</b>           | Engineering Village, One-Search, and IEEE Xplore                                                         | ACM digital library, Google Scholar, IEEE Xplore, ScienceDirect, Web of Science                                                                                    | Google Scholar, IEEE, ACM                                                                                                                                                                                                   |
| <b>Criteria evaluated</b> | Visual impairment; tone; gender; linguistic choice; situation awareness; and trust, privacy and security | Article origin; article type and design; participants; accessibility; metrics; experimental procedure and analysis; automation level; AV details; object detection | Usage of In-Vehicle System; quality of service; user preferences; human-machine interface, driver adaptation and handover mechanisms; ethics, AV acceptance and user perception; inclusion technology and policy frameworks |
| <b>Search strings</b>     | Syntax “(voice assistant) AND (autonomous vehicles)”                                                     | Boolean search term e.g., “head up display” OR “augmented reality”) AND (“autonomous vehicle” OR “self driving vehicle”)                                           | Keywords ‘autonomous vehicles’, ‘accessibility’, ‘inclusion’, and ‘human-computer interaction’                                                                                                                              |
| <b>Initial result</b>     | 253, 3 subsets                                                                                           | 5403                                                                                                                                                               | Over 1300                                                                                                                                                                                                                   |
| <b>Articles analyzed</b>  | 35                                                                                                       | 31                                                                                                                                                                 | Over 100                                                                                                                                                                                                                    |
| <b>Analysis</b>           | PRISMA                                                                                                   | Modified PRISMA                                                                                                                                                    | PRISMA, multi-stage literature screening                                                                                                                                                                                    |

awareness through AR, the review highlighted that very few studies have specifically addressed accessibility for users with cognitive and perceptual limitations.

Bastola et al [6] provided a comprehensive review, analyzing over 100 publications from 2017 to 2024 on AI-powered accessibility enhancements for people with disabilities in AVs. The paper categorized various assistive technologies such as adaptive interfaces, real-time environmental sensing, and inclusion-oriented policy frameworks. This review also pointed out the fragmented focus in existing research, with most studies addressing isolated impairments and lacking integrative, user-centered design approaches.

## 2.4 Towards Inclusive AV Design

Beyond individual studies, an analysis of HCI in mobility revealed that mobility services are often developed for a narrowly defined target group, leading to a lack of inclusive HCI solutions [35]. This narrow focus risks excluding users whose needs do not conform to the predefined categories, especially those with complex or overlapping limitations. Developing inclusive mobility requires solutions that simultaneously address a broad range of physical, cognitive, sensory, and social limitations. Inclusive approaches that accommodate a broader range of users and emphasize ethical consideration are required for the development of current and future mobility systems, including AVs. Without such an inclusive approach, the exclusion of users whose needs fall outside predefined categories is inevitable.

## 3 Methodology

The systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline [43]. The review examines current research at the intersection of HCI, accessibility, and AVs, focusing on how HCI principles are applied to promote accessibility in the context of AVs.

### 3.1 Search Strategy

The literature search was conducted on four academic databases: ACM Digital Library, Scopus, IEEE Xplore, and Web of Science (WOS). These databases were selected due to comprehensive coverage of the integrated topics as well as institutional access availability. A combination of boolean operators and keyword strings related to the main topics of the study was applied to the search:

- **Search string 1**  
"human-computer interaction" OR "Human-machine interaction" OR "Human-technology interaction" OR "Human-centered computing" AND
- **Search string 2**  
"accessibility" OR "Inclusive design" OR "universal design" OR "assistive technology" OR "barrier-free design" AND
- **Search string 3**  
"autonomous vehicle" OR "autonomous bus" OR "automated vehicle" OR "self driving car" OR "automated driving system" OR "intelligent vehicle" OR "autonomous driving" OR "smart vehicle" OR "vehicle automation"

The search strategy was designed to cover variations in terminology used across disciplines. Broader terms related to HCI, accessibility, and AV were intentionally included to account for differences in how research is labeled in the literature. This approach prevents the exclusion of studies whose findings, while not explicitly labeled under a single term, are directly relevant and applicable to the development of AV and accessibility solutions within the HCI scope.

The search strings were adjusted for each database according to the formats. The search was limited to research articles, conference papers, and journal publications published in English between 2015 and 2025, as studies from 2015 onward cover the recent advancements in AV development.

### 3.2 Screening and Selection Process

The initial search identified a total of 536 records from ACM Digital Library (233), Scopus (23), IEEE Xplore (276) and WOS (4). Following the removal of 7 duplicate records and 73 records removed for other reasons such as cover pages, indexes, conference programs, forewords, program booklets, and other administrative documents, a total of 456 records proceeded for title and abstract screening.

In the screening phase, abstracts and titles of each record were evaluated to determine their relevance to the topics. A total of 318 records were excluded for being partially irrelevant or addressing fewer than two of the key topics. An additional 81 records were removed for being completely irrelevant or for insufficiently addressing any of the main topics. After screening, 57 articles were identified as eligible for full-text analysis. Subsequently, 16 publications were excluded for lacking accessibility discussion, 4 for not addressing AVs, and 2 for not being empirical studies. Furthermore, 3 literature reviews were excluded from the final selection but are discussed in Section 2 and summarized in Table 1.

Following this assessment, 32 studies met the inclusion criteria and were selected for the final analysis, which was conducted only on records covering the intersection of HCI, accessibility, and AVs, the primary focus of this paper. The process of identification, screening, and inclusion is illustrated in Figure 2, following the PRISMA flow diagram.

### 3.3 Qualitative Coding

In the next step, the full text content of the 32 selected articles was qualitatively coded in a two-stage process.

First, relevant content was extracted for the following variables:

- Vehicle autonomy - Different types of vehicle autonomy, which are addressed in the scenarios of research studies, are categorized in the context of the Society of Automotive Engineers (SAE) levels [48].
- Target group - Publications are analyzed for their focus on users with special needs, such as persons with visual or physical restrictions, or persons with general age-related reduction in abilities.
- User tasks - It is analyzed which tasks of drivers or passengers have been examined in the study.
- Input interaction mode - The HCI input mode analyzed the modality of input interactions such as voice or touch interactions.
- Output feedback mode - The output mode analyzed the modality of informative feedback of the system to the user.
- Method of usage-centered evaluation - This variable analyzed the type of applied research methods to incorporate user perspectives, for example, interviews, lab experiments or expert-based walkthroughs.

Second, the extracted contents are normalized by eliminating redundancy and grouped into consistent codes. The classification of vehicle autonomy includes the codes *automated* and *semi-autonomous driving*, *fully autonomous driving*, *autonomous shared ride*, and *unspecified autonomy*. The group of automated and semi-autonomous driving comprises scenarios with vehicles up to SAE level 3 "Conditional driving automation" [48], which still require the release of individual driving actions or the handover of the driving action by a

driver. There are two groups for fully autonomous vehicles. On the one hand, fully autonomous driving includes scenarios for individual AV usage with SAE level 4 and 5 for "high driving automation" and "full driving automation" [48]. On the other hand, autonomous shared rides include scenarios with fully autonomous vehicles in a shared or public usage context. Furthermore, all publications that did not address a specified level of vehicle autonomy are coded as *unspecified autonomy*.

The classification of target groups combines restriction-related characteristics with demographic-related characteristics. All physical mobility restrictions that might impact the speed, accuracy or path capability of users are included in *motoric restrictions*, for example, wheelchair users or people with motoric-related diseases such as Parkinson. *Visual restrictions* include publications considering people with blind or low vision. Papers focusing on *elderly people* in general consider slight deficiencies in awareness, reaction times, and special attitudes about AV technology. Publications focusing on more severe *cognitive restrictions* are explicitly coded as such. No publications were found that address particularly *hearing restrictions* or *speech restrictions*. Finally, publications on accessibility issues without addressing a specific restriction are coded as *unspecified target group*.

The analysis of the HCI modes differentiates between the input interaction modes, which describe the way of inserting information and control sequences from the user, and the output feedback modes, which describe the modality of information given by the system to the user. Both include tactile modes for *touch interaction* and *haptic feedback*, *auditory modes* for voice interaction and auditory feedback, and a *group for unspecified modalities*. The input interaction modes also include *gesture* and *brain-computer interfaces*. The output feedback mode groups all screen-based feedback as visual output.

The categories of user tasks grouped several numbers of subtasks according to their relationship to AV. *Driving-related* tasks refer to activities that directly control the driving speed or direction of the vehicle, such as steering, accelerating, or braking. In addition, *vehicle-related* tasks interact or control the status of separate elements of the vehicle that do not influence the driving action, such as controls of the window, door, or climate system. The group of *passenger-related* tasks includes activities for booking, identifying, boarding, and exiting, as well as the local orientation of passengers during their travel without influencing vehicle functions. In addition, tasks for the outside communication process include *interaction of other road users* with special needs with AV. Emergency activities in the meaning of handover tasks are coded as driving-related tasks. Other *emergency-related* tasks refer to calling options for help or other assistance functions in the event of an emergency.

The applied methods of usage-centered evaluation are differentiated into six groups. Experimental studies and usability tests are classified according to their study environment in *lab experiment* and *field studies*. The exploratory methods were differentiated into one group for *guideline-based interviews*, *focus groups*, as well as *surveys*, and one group for creation-oriented *workshops* and *participatory design* processes. Furthermore, studies with the participation of *experts* instead of representative users are coded as expert evaluation and publications conducting an evaluation neither with users nor with experts are coded as *no usage-centered evaluation*.

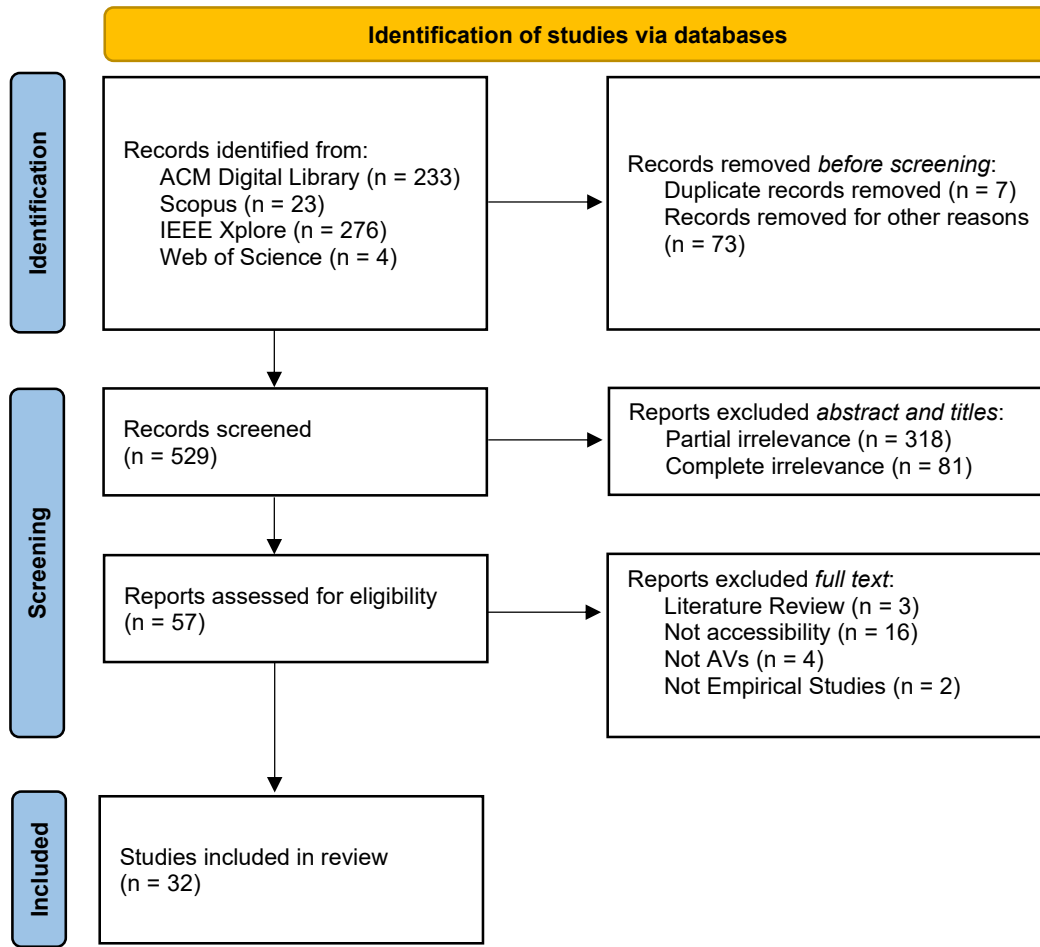


Figure 2: PRISMA flow diagram for systematic literature review.

## 4 Results

The quantitative distribution of the coding results is shown in Table 2. In the following, these results are analyzed with regard to their distribution within the category of vehicle autonomy. Since the four vehicle autonomy groups contain a different total number of publications analyzed, the results in the following chapters are compared based on their relative percentages.

### 4.1 Target Group

Figure 3 shows that people with visual limitations are the group that was considered the most frequently in the analyzed studies (12 publications). Particularly in the area of fully AVs with individual use, users with visual restrictions comprised the majority of the targeted groups (64 percent). Elderly persons in general are the group that is considered second most frequently. In the field of automated and semi-autonomous driving, elderly persons represent the focus of accessibility considerations, without addressing specific limitations. The proportion of studies without reference to specific target groups from the area of users with special needs

is high in the areas of automated and semi-autonomous driving (40 percent) and unspecified vehicle autonomy (36 percent). Only the autonomous shared ride group considered four different target groups with restrictions, including a study that considered children with cognitive limitations. Accessibility for people with motor restrictions is analyzed in small numbers for autonomous shared rides, fully autonomous driving, and studies with unspecified vehicle autonomy. Furthermore, the analysis also included restrictions on hearing and speech capabilities, but without any results.

### 4.2 User Tasks

In 14 of 32 studies, no specific user tasks are addressed. This large number of studies primarily analyzes the acceptance, attitudes, and expectations of passengers with special needs about possible HCI with AVs without addressing specific interaction processes. The user tasks most frequently considered are passenger-related tasks (10 publications), which, in addition to booking, identifying, boarding, and exiting processes, also deal with the passive orientation of passengers during the journey. As shown in Figure 4, these tasks

**Table 2: Overview of categories and codes resulting of qualitative analysis**

| Categories and codes             | Selected | Percentage | Publications                                               |
|----------------------------------|----------|------------|------------------------------------------------------------|
| <b>Vehicle autonomy</b>          |          |            |                                                            |
| Automated/semi-autonomous        | 5        | 15.6       | [7, 28, 46, 47, 51]                                        |
| Fully autonomous driving         | 11       | 34.4       | [1, 3, 4, 13, 17, 20–22, 33, 37, 39]                       |
| Autonomous shared ride           | 5        | 15.6       | [2, 12, 24, 34, 54]                                        |
| Unspecified autonomy             | 11       | 34.4       | [9–11, 14, 18, 25, 27, 32, 36, 38, 52]                     |
| <b>Target group</b>              |          |            |                                                            |
| Motor restriction                | 5        | 15.6       | [4, 27, 34, 39, 54]                                        |
| Visual restriction               | 12       | 37.5       | [10–13, 17, 20–22, 33, 34, 37, 52]                         |
| Cognitive restriction            | 1        | 3.1        | [2]                                                        |
| Elderly persons                  | 9        | 28.1       | [3, 7, 9, 24, 25, 28, 32, 39, 51]                          |
| Unspecified target group         | 7        | 21.9       | [1, 14, 18, 36, 38, 46, 47]                                |
| <b>User tasks</b>                |          |            |                                                            |
| Driving-related tasks            | 8        | 25.0       | [3, 7, 10, 21, 27, 46, 47, 51]                             |
| Vehicle-related tasks            | 5        | 15.6       | [7, 10, 11, 34, 39]                                        |
| Passenger-related tasks          | 10       | 31.3       | [3, 11–13, 20, 32, 34, 37, 39, 54]                         |
| Communicating from outside       | 4        | 12.5       | [4, 13, 17, 52]                                            |
| Emergency-related tasks          | 2        | 6.3        | [34, 54]                                                   |
| Unspecified user tasks           | 14       | 43.8       | [1, 2, 7, 9, 14, 18, 22, 24, 25, 28, 32, 33, 36, 38]       |
| <b>Input interaction mode</b>    |          |            |                                                            |
| Tactile/touch                    | 11       | 34.4       | [3, 4, 13, 32–34, 37, 39, 51, 52, 54]                      |
| Gesture                          | 2        | 6.3        | [21, 33]                                                   |
| Voice                            | 9        | 28.1       | [3, 10, 11, 13, 17, 22, 33, 34, 52]                        |
| Brain-computer-interface         | 1        | 3.1        | [27]                                                       |
| Unspecified input interaction    | 14       | 43.8       | [1, 2, 7, 9, 12, 14, 18, 24, 25, 28, 36, 38, 46, 47]       |
| <b>Output feedback mode</b>      |          |            |                                                            |
| Tactile                          | 7        | 21.9       | [4, 10, 17, 20–22, 37]                                     |
| Visual                           | 7        | 21.9       | [4, 7, 32, 34, 39, 51, 54]                                 |
| Auditory                         | 5        | 15.6       | [4, 11, 21, 34, 37]                                        |
| Unspecified feedback             | 15       | 46.9       | [1, 9, 12, 14, 18, 24, 25, 27, 28, 33, 36, 38, 46, 47, 52] |
| <b>Usage-centered evaluation</b> |          |            |                                                            |
| Lab experiment                   | 9        | 28.1       | [3, 17, 20, 21, 27, 32, 37, 39, 51]                        |
| Field study                      | 2        | 6.3        | [2, 12]                                                    |
| Interview / focus group / survey | 10       | 31.3       | [2, 3, 7, 10, 13, 18, 21, 28, 34, 54]                      |
| Workshop / participatory design  | 7        | 21.9       | [2, 4, 10, 17, 24, 34, 37]                                 |
| Expert evaluation                | 3        | 9.4        | [12, 18, 22]                                               |
| No user-centered evaluation      | 12       | 37.5       | [1, 9, 11, 14, 18, 25, 33, 36, 38, 46, 47, 52]             |

are especially relevant in fully autonomous and shared autonomous cases and represent the majority of tasks in both groups. In contrast, studies for automated and semi-autonomous driving focus on driving-related tasks for handover situations, including steering, accelerating, and braking. Some driving-related tasks are also considered for fully autonomous driving situations, as requested by passengers. In addition, vehicle-related tasks, such as door or heating controls, are represented in studies in the context of all vehicle autonomy groups. Some studies also refer to the communication process of other road users with special needs with AV.

Emergency-related tasks are only considered in two publications for autonomous shared rides.

### 4.3 Human-Computer Interaction

Most of the analyzed publications did not specify the input or output modality, and 12 have even specified neither of them. While touch interactions are the most investigated modality and are considered in all groups, voice interactions are mainly analyzed for the full and shared vehicle autonomy level, as shown in Figure 5. In addition, gesture-based interaction and brain-computer interfaces are only considered in isolated cases. Five publications applied multimodal

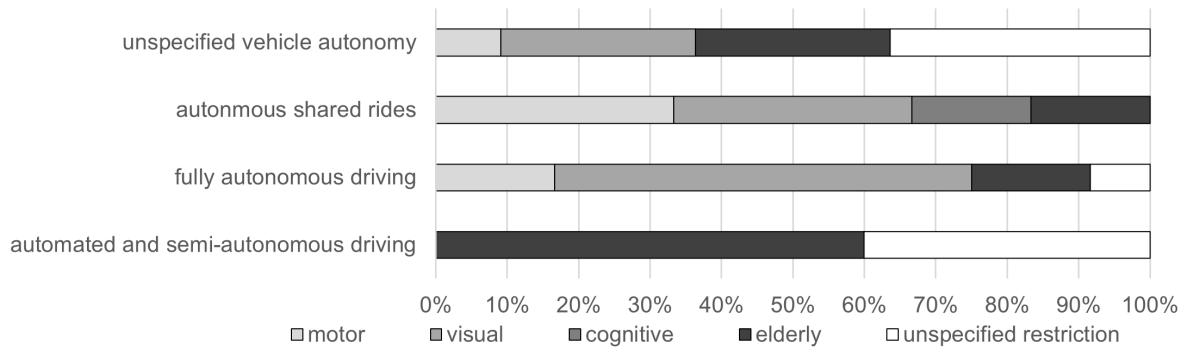


Figure 3: Percentage of target groups grouped by vehicle autonomy (n=32)

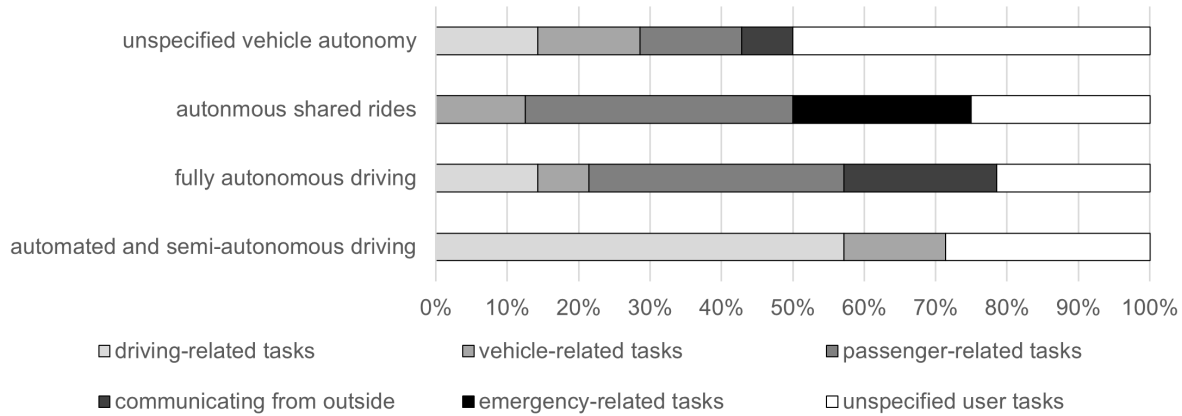


Figure 4: Percentage of tasks grouped by vehicle autonomy (n=32)

interaction approaches, exemplarily combining touch and voice interaction.

The distribution of the three coded output modalities, tactile, visual, and auditory, extends across all vehicle autonomy groups, as shown in Figure 6. Only the automated and semi-autonomous driving group focuses on visual feedback, while tactile feedback is considered in particular for fully autonomous driving.

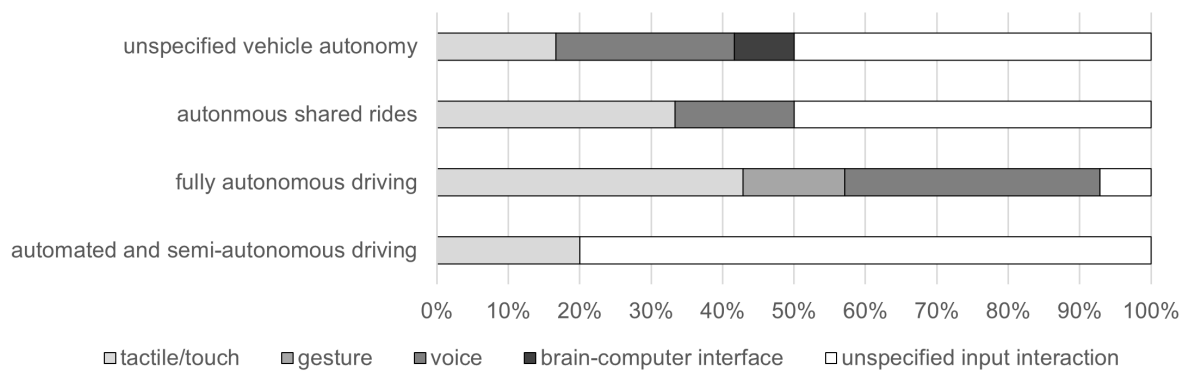
#### 4.4 Usage-centered Evaluation

Only 12 of the analyzed publications contain a use-oriented technological solution proposal, a usage-centered proposal for a design, or an evaluation process that is not itself assessed by a usage-oriented evaluation. Figure 7 shows that the most frequently applied evaluation method for accessibility issues for AV are personal questioning and discussion in interviews, focus groups, or surveys, as applied in all contexts of AV. Studies in the context of fully and shared autonomy used workshops or other methods of participatory design just as frequently as interviews, focus groups, and surveys. This high proportion of open and exploratory methods provides general insights into attitudes and acceptance for the introduction

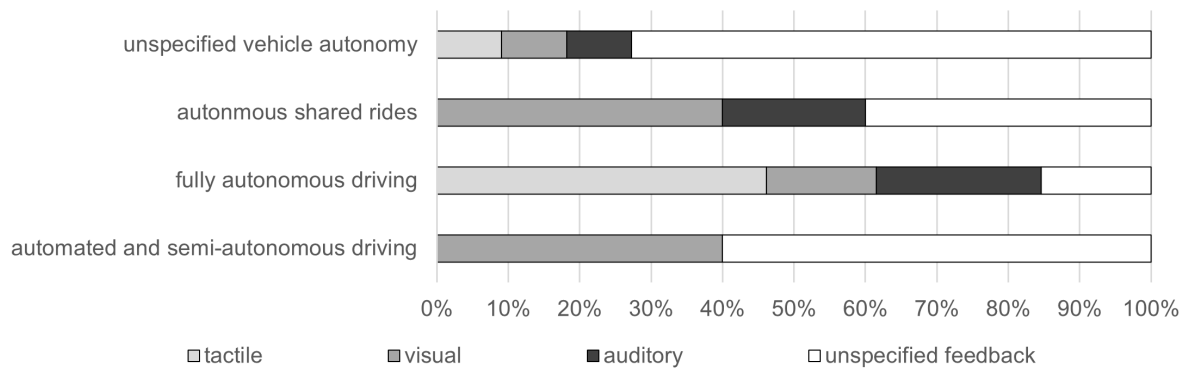
of technology, but lacks practical usage contexts. In addition, expert evaluations are applied in a small number to compensate for the lack of available fields of practical implementation. Lab experiments are especially applied for future scenarios of fully autonomous individual driving, and two field studies considered already implemented autonomous shared rides.

#### 5 Discussion and Research Gaps

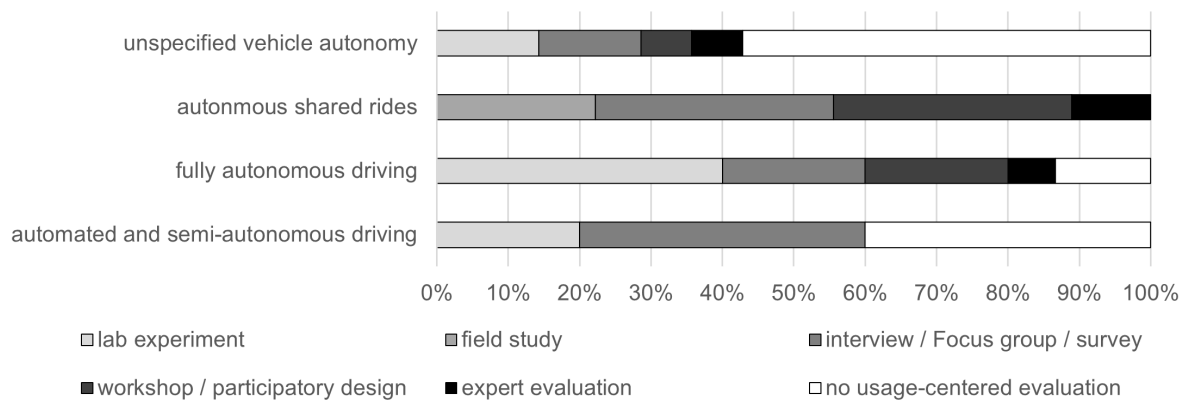
The qualitative analysis related to the target group in this study showed research on accessibility in AV has mainly focused on visual restrictions and elderly populations, while significantly less attention has been paid to other groups such as hearing, motor, and cognitive restrictions. Such an imbalance risks creating AV systems that exclude large segments of users. This finding aligns with the prior study by Chu and Huang [16], who emphasized the role of voice assistants in supporting users with visual and motor impairments in AV contexts. The consideration of various target groups of users with different special needs within autonomous shared rides emphasizes the particular importance and potential of



**Figure 5: Percentage of input interaction modes grouped by vehicle autonomy (n=32)**



**Figure 6: Percentage of output feedback modes grouped by vehicle autonomy (n=32)**



**Figure 7: Percentage of usage-centered evaluation methods focused grouped by vehicle autonomy (n=32)**

this level of vehicle autonomy for mobility that is accessible to all users with special needs.

Although technologies such as vibrotactiles and integration with hearing devices have been explored to support communication,

their application within AVs remains limited for users with hearing impairments. The analysis shows that input and output modalities are often unspecified in existing studies, with touch-based interaction being the most frequently considered modality. However, touch



interfaces may pose challenges for users with motor impairments. Despite the recognized potential of voice interaction to enhance accessibility, its implementation is mostly limited to users with visual impairments. In general, the growing demand for multimodal communication, which combines visual, auditory, and tactile feedback, has shown potential benefits for users with special needs. However, these approaches also reveal usability challenges, including risk of overwhelming users with excessive information, conflicts between modalities and user abilities, and the lack of interfaces that adapt to individual needs[45].

The results related to the usage-oriented methods show that research on accessible HCI with AVs is focused on open and exploratory methods such as workshops, focus groups, and interviews. There is still a gap in various practical usage contexts for field studies with AVs and users with special needs, often caused by legal and organizational limitations. In addition, HCI in emergency situations for AVs remains an area that requires further investigation, particularly in practical contexts, to support effective and inclusive system responses for users with special needs.

Finally, usage-oriented studies in the field of accessibility and AVs still focus on tasks oriented towards individual driving behavior or shared travel behavior. Public transportation contexts and emergency contexts are given too little consideration.

## 6 Conclusion

AVs have the potential to transform mobility by shifting tasks from humans to automated systems, particularly benefiting individuals who face barriers to driving, such as older adults and people with disabilities. As AV technologies advanced towards widespread adoption, addressing accessibility through the perspective of HCI becomes increasingly vital. This literature review highlighted the importance of integrating accessibility considerations into AVs development by analyzing how current research addresses the needs of users with special needs. Although some efforts focus on older adults and individuals with visual impairments, substantial gaps remain in addressing the requirements of users with hearing and cognitive impairments. Future work should consider participatory design methods and expand the inclusion of underrepresented groups to support the development of AV systems that are accessible and inclusive to diverse user needs.

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